

LAB 12

Task # 01, 02, 03, 04, 05:

```
import java.io.File;
import java.io.FileNotFoundException;
import java.io.FileWriter;
import java.io.IOException;
import java.util.Scanner;
import java.io.FileInputStream;
import java.io.FileOutputStream;

class Main{
    public static void main(String[] args){
        String filename = "readme.txt";
        createFile(filename);
        writeToFile(filename, "Hello! I am muneeb.");
        appendText(filename, "After appending to an existing file");
        Readfile(filename);
        getFileInfo(filename);
        FileIOStream(filename, "output.txt");
        deleteFile(filename);
    }
    static void createFile(String filename) {
        File file = new File(filename);
        try{
            file.createNewFile();
            System.out.println("File created successfully");
        } catch (IOException e) {
            throw new RuntimeException(e);
        }
    }

    static void writeToFile(String filename, String content) {
        try{
            FileWriter fw = new FileWriter(filename);
            fw.write(content);
            fw.close();
            System.out.println("Wrote to " + filename);
        } catch (IOException e) {
            System.out.println("Error writing to " + filename);
        }
    }

    static void Readfile(String filename) {
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try{
    File file = new File(filename);
    Scanner myReader = new Scanner(file);
    System.out.println(myReader.nextLine());
    while (myReader.hasNextLine()) {
        System.out.println("Reading File");
        String text = myReader.nextLine();
        System.out.println(text);
    }
    myReader.close();
}
catch (FileNotFoundException e) {
    System.out.println("Error reading from " + filename);
}
}

static void getFileInfo(String filename){
    File file = new File(filename);
    if(file.exists()){
        System.out.println("File exists");
        System.out.println("File name: " + file.getName());
        System.out.println("Absolute path: " + file.getAbsolutePath());
        System.out.println("Writeable: " + file.canWrite());
        System.out.println("Readable: " + file.canRead());
        System.out.println("File size: " + file.length());
    }else{
        System.out.println("File does not exist");
    }
}

static void appendText (String filename, String content){
    try (FileWriter writer = new FileWriter(filename, true)) {
        writer.write(content);
    } catch (IOException e) {
        System.err.println("Error writing to file: " + e.getMessage());
    }
}

static void deleteFile(String filename){
    File file = new File(filename);
    if(file.exists()){
        file.delete();
        System.out.println("File deleted successfully");
    }
}
```

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}

static void FileIOStream(String inputFile, String outputFile){
    FileInputStream fileInputStream = null;
    FileOutputStream fileOutputStream = null;

    try {
        // Create FileInputStream to read from the input file
        fileInputStream = new FileInputStream(inputFile);

        // Create FileOutputStream to write to the output file
        fileOutputStream = new FileOutputStream(outputFile);

        int byteData;
        // Read byte by byte from the input file and write to the output file
        while ((byteData = fileInputStream.read()) != -1) {
            fileOutputStream.write(byteData);
        }

        System.out.println("File has been copied successfully.");
    } catch (IOException e) {
        // Handle potential IOException
        System.err.println("An IOException occurred: " + e.getMessage());
    } finally {
        try {
            // Close the streams if they are not null
            if (fileInputStream != null) {
                fileInputStream.close();
            }
            if (fileOutputStream != null) {
                fileOutputStream.close();
            }
        } catch (IOException e) {
            System.err.println("Error closing streams: " + e.getMessage());
        }
    }
}
```